🎯 What is Datadog?

Datadog is a cloud-based **monitoring and analytics platform** for IT operations and development teams. It helps track the health and performance of applications, infrastructure, logs, and more.

🛠️ 1. Installing the Datadog Agent

🔑 Prerequisites:

- A Datadog account: https://www.datadoghq.com
- Access to the system (Linux/Windows/macOS) where you want to install the agent
- Datadog API Key (from your Datadog dashboard)

🐧 On Linux (Debian/Ubuntu)

```
bash

DD_API_KEY=<YOUR_DATADOG_API_KEY> bash -c "$(curl -L
https://s3.amazonaws.com/dd-agent/scripts/install_script.sh)"
```

🍎 On macOS (using Homebrew)

```
bash

brew install --cask datadog-agent
```

Then edit the configuration:

```
bash

sudo vi /opt/datadog-agent/etc/datadog.yaml
# Add your API key
```

🪟 On Windows

1. Download the installer from your Datadog dashboard.
2. Run it and provide the API key during setup.

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📊 2. Types of Metrics in Datadog

Datadog supports **four metric types**:

Type	Description	Examples
Gauge	Represents a single value at a point in time	CPU usage, memory usage
Count	Tracks the number of times an event occurs	Number of requests
Rate	Count per second over a time interval (used for throughput)	Requests/sec
Histogram	Measures the statistical distribution of a set of values	Request latency distribution
Distribution	Like Histogram, but aggregates across hosts and services for global stats	Global request latency

✅ **Tip:** Use tags (like `env:prod`, `region:us-east`) to categorize and filter metrics.

⚙️ 3. Key Components of Datadog

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3.1 Dashboards

- Visual interface to display metrics and logs
- You can add graphs, heatmaps, tables, etc.

3.2 Monitors

- Alerts based on metrics, logs, or events
- Types:
  - Metric alert
  - Anomaly detection
  - Forecast alerts
  - Composite alerts (combine multiple conditions)

3.3 Integrations

- Pre-built integrations for AWS, Azure, GCP, Kubernetes, Docker, etc.
- Easily collect metrics without extra code

3.4 Tags

- Used to filter, group, and search data
- Examples: `service:web`, `env:prod`, `team:backend`

🔍 4. Logs, Traces, and APM

Logs

- Install log collection with the Datadog Agent
- Configure `/etc/datadog-agent/conf.d/<your_service>.d/conf.yaml`
- Logs can be parsed and searched in real time

Traces (APM - Application Performance Monitoring)

- Trace HTTP requests, database queries, etc.
- Supported in languages like Python, Java, Node.js, Ruby, Go

Example for Python (with `ddtrace`):

```
bash

pip install ddtrace
python

from ddtrace import tracer

@tracer.wrap()
def process_data():
    # your logic here
```

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📚 5. Hands-On Activities (for students)

Activity 1: Install the Datadog agent on a VM (Linux or macOS)

Activity 2: Monitor CPU, memory, and disk usage

Activity 3: Set up a custom metric (using DogStatsD)

Activity 4: Create a dashboard to visualize metrics

Activity 5: Set up an alert for high CPU usage

🧠 6. Best Practices

- Always tag your metrics and services (`env`, `region`, `service`, `team`)
- Use dashboards to give teams visibility into their own systems
- Enable anomaly detection for smarter alerting
- Set up role-based access control (RBAC) for secure usage
- Integrate with Slack, PagerDuty, or email for alerting

🧩 7. Extra: Custom Metrics with DogStatsD

Datadog supports **DogStatsD**, a metrics aggregation daemon.

Example (Python):

```
python

from datadog import initialize, statsd

options = {'api_key':'YOUR_API_KEY','app_key':'YOUR_APP_KEY'}
initialize(**options)

statsd.gauge('pathnex.cpu', 75, tags=["env:dev", "app:pathnex"])
```

🐋 8. Monitoring Kubernetes with Datadog

🧰 Prerequisites

- A Kubernetes cluster (EKS, GKE, AKS, Minikube, etc.)
- `kubectl` configured
- Helm installed (recommended)

📦 Installation via Helm (recommended)

```
bash

helm repo add datadog https://helm.datadoghq.com
helm repo update

helm install datadog-agent datadog/datadog \
  --set datadog.apiKey=<YOUR_DATADOG_API_KEY> \
  --set datadog.site='datadoghq.com' \
  --set targetSystem=linux
```

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🔍 Key Features

- Autodiscovery of Kubernetes services
- Container-level metrics (CPU, memory, restarts)
- Cluster health overview
- Live container view
- Kubelet & control plane monitoring
- DaemonSet deployment ensures each node is covered

✅ Best Practices

- Enable resource tagging (`env`, `team`, `app`)
- Use Datadog's Kubernetes integration dashboard
- Monitor pod restarts, OOM kills, and node status
- Set up monitors for failed deployments and crash loops

🧪 9. Monitoring Apache Kafka with Datadog

📦 Prerequisites

- Datadog Agent running on the same host as Kafka brokers
- Java/JMX metrics enabled on Kafka

🗂️ Install Kafka Integration

1. Enable the JMX metrics in Kafka config (`KAFKA_JMX_OPTS`)
2. Modify or create:
   `/etc/datadog-agent/conf.d/kafka.d/conf.yaml`

```
yaml

init_config:
  is_jmx: true

instances:
  - host: localhost
    port: 9999
    name: kafka-local
    tags:
      - env:dev
      - service:kafka
```

3. Restart the Datadog Agent:

```
bash

sudo service datadog-agent restart
```

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📊 Metrics Collected

- Broker: `kafka.server.BrokerTopicMetrics.*`
- Consumer: `kafka.consumer.*`
- Producer: `kafka.producer.*`
- Topics/Partitions: lag, throughput, error rates

✅ Best Practices

- Tag metrics by topic, consumer group, and environment
- Set alerts for high consumer lag and under-replicated partitions
- Use the Datadog Kafka dashboard template to get started

Summary Table

Topic	Key Takeaway
Installation	Setup agent via script or Helm
Kubernetes	Use Helm for agent deployment; cluster-wide
Kafka	Use JMX with the Agent for detailed metrics
Metric Types	Gauge, Count, Rate, Histogram, Distribution
Dashboards	Graphs for CPU, memory, logs, APM, etc.
Monitors	Alert on performance and anomalies
APM & Tracing	Trace requests across services and layers
Custom Metrics	Use DogStatsD or client libraries
Integrations	Plug into services like AWS, Docker, etc.

Important links:

https://docs.datadoghq.com/getting_started/agent/

https://docs.datadoghq.com/monitors/types/

https://docs.datadoghq.com/getting_started/application/

https://docs.datadoghq.com/getting_started/monitors/

https://docs.datadoghq.com/getting_started/logs/

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